DRUHÉ PARCIÁLNE DERIVÁCIE

$f(x,y) \rightarrow f'_x, f'_y \rightarrow f''_{xx}, f''_{xy}, f''_{yy}, f''_{yx}$

Príklad $f(x,y) = x^3 - 3x^4y + y^5$

y = konštanta

$f'_x = 3x^2 - 12x^3y = 3x^2 - 12x^3y$

$f'_y = -3x^4 \cdot 1 + 5y^4 = -3x^4 + 5y^4$

y = konštanta

$f''_{xx} = 6x - 36x^2y$

x = konštanta

$f''_{xy} = -12x^3$

y = konštanta

$f''_{yx} = -12x^3$

x = konštanta

$f''_{yy} = 20y^3$

rovnaké

Príklad $f(x,y) = xy + \frac{y}{x}$

y = konštanta

$f'_x = y + \frac{-y}{x^2} = y - \frac{y}{x^2}$

x = konštanta

$f'_y = x + \frac{x}{x^2} = x + \frac{1}{x}$

y = konštanta

$f''_{xx} = 0 - \frac{-y \cdot 2x}{x^4} = \frac{2yx}{x^4} = \frac{2y}{x^3}$

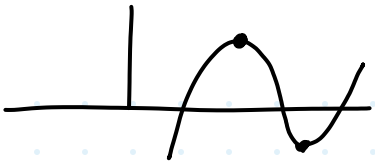
x = konštanta

$f''_{xy} = 1 - \frac{1 \cdot x^2 - 0}{x^4} = 1 - \frac{x^2}{x^4} = 1 - \frac{1}{x^2}$

x = konštanta

$f''_{yy} = 0 + \frac{0}{x^2} = 0$

LOKÁLNE EXTRÉMY



↳ lokálny extrém funkcie je bod, v ktorom funkcia nadobúda svoju najmenšiu (lokálne minimum) alebo najväčšiu (lokálne maximum) hodnotu v porovnaní s bezprostredným okolím tohto bodu.

Zadanie: Nájdite lokálne extrémy funkcie

KROKY:

1) Prvé parciálne derivácie $f(x,y)$

$$\left.\begin{aligned} f'_x &= 0 \\ f'_y &= 0 \end{aligned}\right\}$$ dve rovnice o dvoch neznámych

$\Rightarrow$ vyšlo body, budú kandidáti na extrémy

2) Hessián

$$\begin{vmatrix} f''_{xx} & f''_{xy} \\ f''_{yx} & f''_{yy} \end{vmatrix} = f''_{xx} \cdot f''_{yy} - f''_{xy} \cdot f''_{yx}$$

3) Body (kandidáti) dosadíme do Hessiánu = ČÍSLO

$A[0,0]$

ČÍSLO > 0 MÁME EXTRÉM
< 0 NEMÁME EXTRÉM
$= 0$ NEVIEME VEDIEŤ

f''_{xx} > 0 MINIMUM
< 0 MAXIMUM

Príklad Nájdite lokálne extrémy funkcie $f(x,y) = 6xy - x^2y - xy^2$

1) Prvé parciálne derivácie :

$$f'_x = 6y - 2xy - y^2 = 0 \quad (1)$$

$$f'_y = 6x - x^2 - 2xy = 0 \quad (2)$$

(1) $6y - 2xy - y^2 = 0$

$y(6 - 2x - y) = 0$

$y = 0$ dosadíme do (2)

$6x - x^2 - 2x \cdot 0 = 0$

$6x - x^2 = 0$

$x(6 - x) = 0$

$x = 0$ $\quad$ $x = 6$

$A[0,0]$ $\quad$ $B[6,0]$

$6 - 2x - y = 0$

$6 - 2x = y$

$y = 6 - 2x = 6 - 2 \cdot 0 = 6$

dosadím do (2)

$6x - x^2 - 2x(6 - 2x) = 0$

$6x - x^2 - 12x + 4x^2 = 0$

$3x^2 - 6x = 0 \quad / 3$

$x^2 - 2x = 0$

$x(x - 2) = 0$

$x = 0$ $\quad$ $x = 2$

$y = 6 - 2x = 6 - 2 \cdot 2 = 2$

$C[0,6]$ $\quad$ $D[2,2]$

2) Hessián

$$\begin{vmatrix} f''_{xx} & f''_{xy} \\ f''_{yx} & f''_{yy} \end{vmatrix}$$

$f''_{xx} = -2y$

$f''_{xy} = 6 - 2x - 2y$

$f''_{yy} = -2x$

$f'_x = 6y - 2xy - y^2$

$f'_y = 6x - x^2 - 2xy$

$$\begin{vmatrix} -2y & 6-2x-2y \\ 6-2x-2y & -2x \end{vmatrix} = (-2y)\cdot(-2x) - (6-2x-2y)^2$$

3) Dosadíme nájdené body do Hessiánu

$A[0,0] = (-2\cdot 0)\cdot(-2\cdot 0) - (6 - 2\cdot 0 - 2\cdot 0)^2 = -36 < 0$ NEMÁM EXTRÉM

$B[6,0] = (-2\cdot 0)\cdot(-2\cdot 6) - (6 - 2\cdot 6 - 2\cdot 0)^2 = -36 < 0$ X

$C[0,6] = (-2\cdot 6)\cdot(-2\cdot 0) - (6 - 2\cdot 0 - 2\cdot 6)^2 = -36 < 0$ X

$D[2,2] = (-2\cdot 2)\cdot(-2\cdot 2) - (6 - 2\cdot 2 - 2\cdot 2)^2 = 12 > 0$ MÁM EXTRÉM

$D[2,2]$

$f''_{xx} = -2y = -2 \cdot 2 = -4 < 0$ MAXIMUM

Funkcia $f(x,y)$ má v bode $[2,2]$ lokálne maximum s hodnotou 8.

$[2,2] \rightarrow f(x,y) = 6xy - x^2y - xy^2 = 6 \cdot 2 \cdot 2 - 2^2 \cdot 2 - 2 \cdot 2^2 =$
$24 - 8 - 8 = 8$

Príklad Nájdite lokálne extrémy funkcie $f(x,y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$

1) PPD

$f'_x = 3x^2 + 3y^2 - 6x = 0 \quad / \cdot 3 \quad x^2 + y^2 - 2x = 0 \quad (1)$

$f'_y = 6xy - 6y = 0 \quad / \cdot 6 \quad xy - y = 0 \quad (2)$

(2) $xy - y = 0$

$y(x-1) = 0$

$y = 0$ dosadíme do (1)

$x^2 + 0^2 - 2x = 0$

$x^2 - 2x = 0$

$x(x-2) = 0$

$x = 0$ → $A[0,0]$

$x = 2$ → $B[2,0]$

$x = 1$ dosadíme do (1)

$1^2 + y^2 - 2 \cdot 1 = 0$

$1 + y^2 - 2 = 0$

$y^2 = 1$

$y = 1$ → $C[1,1]$

$y = -1$ → $D[1,-1]$

2) Hessián

$$\begin{vmatrix} f''_{xx} & f''_{xy} \\ f''_{yx} & f''_{yy} \end{vmatrix} =$$

$f''_{xx} = 6x - 6$

$f''_{xy} = 6y$

$f''_{yy} = 6x - 6$

$f'_x = 3x^2 + 3y^2 - 6x$

$f'_y = 6xy - 6y$

$$\begin{vmatrix} 6x-6 & 6y \\ 6y & 6x-6 \end{vmatrix} = (6x-6)^2 - (6y)^2$$

3) $A[0,0] = (6\cdot 0-6)^2 - (6\cdot 0)^2 = 36$ ✓ MÁME EXTRÉM (+36)

$B[2,0] = (6\cdot 2-6)^2 - (6\cdot 0)^2 = 36$ ✓ MÁME EXTRÉM

$C[1,1] = (6\cdot 1-6)^2 - (6\cdot 1)^2 = -36$ ✗

$D[1,-1] = (6\cdot 1-6)^2 - (6\cdot(-1))^2 = -36$ ✗

$f''_{xx} = 6x-6$ =

$A[0,0] = -6 < 0$ MAXIMUM

$B[2,0] = 6 > 0$ MINIMUM

Funkcia $f(x,y)$ má v bode $[0,0]$ lokálne maximum s hodnotou 4 a v bode $[2,0]$ lokálne minimum s hodnotou 0.

Príklad Nájdite lokálne extrémy funkcie $f(x,y) = 4x^3 + xy^2 + 2x^2 + y^2$

1) $f'_x = 12x^2 + y^2 + 4x = 0$ (1) $\quad 12x^2+y^2+4x=0$

$f'_y = 2xy + 2y = 0 \mid 2$ (2) $\quad xy + y = 0$

(2) $xy + y = 0$

$y(x+1) = 0$

$y = 0$ $\qquad$ $x = -1$

(1) $\qquad$ (1)

$12x^2 + 0^2 + 4x = 0$ $\qquad$ $12\cdot(-1)^2 + y^2 + 4(-1) = 0$

$12x^2 + 4x = 0$ $\qquad$ $12 + y^2 - 4 = 0$

$x(12x+4) = 0$ $\qquad$ $y^2 + 8 = 0$

$y^2 = -8$ ✗

$x = 0$ $\qquad$ $12x = -4$

$x = -\frac{4}{12} = -\frac{1}{3}$

$A[0,0]$ $\qquad$ $B[-\frac{1}{3}, 0]$

2) Hessián

$f''_{xx} = 24x + 4$

$f''_{xy} = 2y$

$f''_{yy} = 2x + 2$

$f'_x = 12x^2 + y^2 + 4x$

$f'_y = 2xy + 2y$

$$\begin{vmatrix} 24x+4 & 2y \\ 2y & 2x+2 \end{vmatrix} = (24x+4)(2x+2) - (2y)^2$$

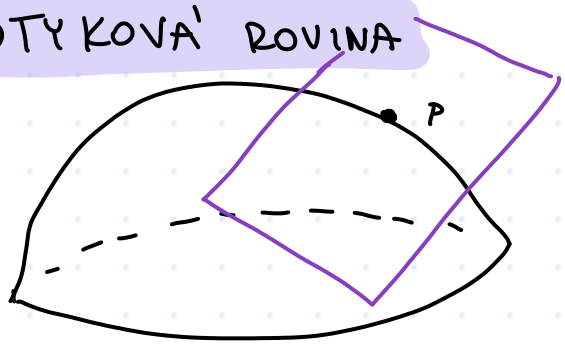
3) $A[0,0] = (24\cdot 0 + 4)(2\cdot 0 + 2) - (2\cdot 0)^2 = 8 > 0$ MÁME EXTRÉM

$B\left[-\frac{1}{3},0\right] = \left(24\cdot\left(-\frac{1}{3}\right) + 4\right)\left(2\cdot\left(-\frac{1}{3}\right) + 2\right) - (2\cdot 0)^2 = -\frac{16}{3} < 0$ ✗ NEMÁME EXTRÉM

$f''_{xx} = 24x + 4 \quad A[0,0] = 24\cdot 0 + 4 = 4 > 0$ MINIMUM

Funkcia $f(x;y)$ má v bode $[0,0]$ lokálne minimum s hodnotou 0

DOTYKOVÁ ROVINA



VŠEOBECNÁ ROVNICA ROVINY V BODE $B[x_0, y_0]$

$$\underbrace{\frac{\partial f}{\partial x}(x_0,y_0)}_{\text{číslo}}(x-x_0)+\underbrace{\frac{\partial f}{\partial y}(x_0,y_0)}_{\text{číslo}}(y-y_0)=z-\underbrace{f(x_0,y_0)}_{\text{číslo}}$$

Príklad $z = 1-\frac{1}{10}(x^2+4y^2)$ v bode $P[\overset{x_0}{1},\overset{y_0}{1},?]$

$$z = 1-\frac{1}{10}(1^2+4\,1^2) = 1-\frac{5}{10} = 1-\frac{1}{2} = \frac{1}{2}$$

$$f'_x = 0-\frac{1}{10}(2x) = -\frac{2x}{10} = -\frac{x}{5} \Rightarrow P[1,1,?] \Rightarrow -\frac{1}{5}$$

$$f'_y = 0-\frac{1}{10}\cdot(8y) = -\frac{8y}{10} = -\frac{4y}{5} \Rightarrow P[1,1,?] \Rightarrow -\frac{4}{5}$$

$$-\frac{1}{5}(x-1)-\frac{4}{5}(y-1) = z-\frac{1}{2}$$

Príklad Nájdite rovnicu dotykovej roviny k funkcii $f(x,y) = x^2y^3+y^3-5$ v bode $P[\overset{x_0}{2},\overset{y_0}{-2},?]$

$$z = 2^2(-2)^3+(-2)^3-5 = -45$$

$$f'_x = 2xy^3 \Rightarrow P[2,-2,?] \Rightarrow 2\cdot2(-2)^3 = -32$$

$$f'_y = 3x^2y^2+3y^2 \Rightarrow P[2,-2,?] \Rightarrow 3\cdot2^2\cdot(-2)^2+3\cdot(-2)^2 = 60$$

$$-32(x-2)+60(y+2) = z+45$$